

Regularized Multivariate Two-way Functional Principal Component Analysis

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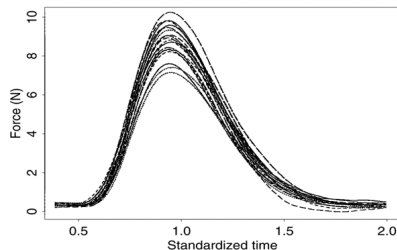
Outline

- 1 Introduction & Background
- 2 A SVD Approach for Regularized Multivariate FPCA
- 3 Two-way Regularized Multivariate FPCA
- 4 Conclusion & Future Work

Introduction

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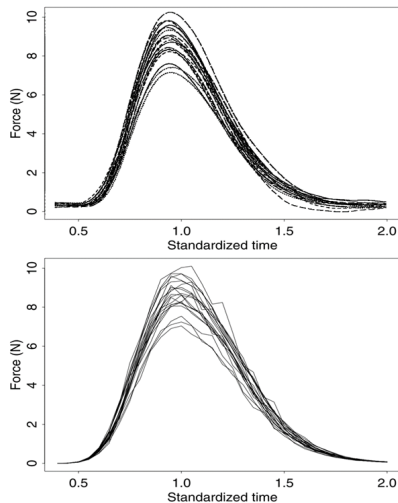
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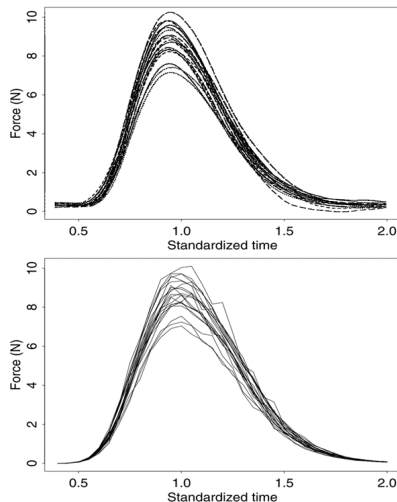
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- These data are often recorded at discrete time points or grid locations, even though they originate from continuous processes in areas like engineering, finance, environmental science, and healthcare.
- Functional Data Analysis (FDA) is a statistical framework that treats these observations as realizations of smooth underlying functions, allowing for more accurate modeling and interpretation of continuous processes.



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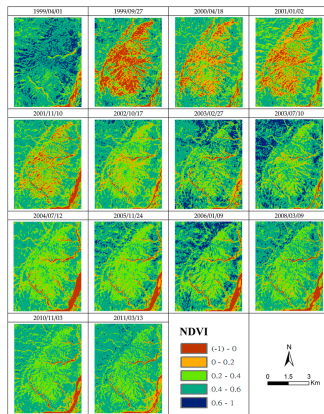
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- Impact: More adaptable, robust, and applicable across diverse scientific and business problems.

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 - Effective for dimension reduction in complex datasets.

Mathematical Foundation of FPCA

- **Goal:** Identify functional directions that maximize variance (low-rank approximation of functional data). For functional data $X \in \mathbb{R}^{n \times m}$ contains the discretized functional observations (rows correspond to subjects, columns to grid points), $v \in \mathbb{R}^m$ represents the estimated principal component (function), and $u \in \mathbb{R}^n$ denotes the associated principal component scores.

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- **Optimization steps:**

$$\text{Fix } v : u = \frac{Xv}{v^\top v} \quad \text{and} \quad \text{Fix } u : v = \frac{X^\top u}{u^\top u}$$

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- Algorithms: Based on iterative **power method** and **thresholding** updates.

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- Consider the SVD of X as $X = UDV^\top$, where U and V have orthonormal columns and D is diagonal with ordered singular values. In particular, for $X = u_d v_d^\top$, v is the first principal component and $u = u_d$ gives the associated scores. With these representations, the power algorithm (described below) converges quickly, typically in only a few iterations.

Power Algorithm

Algorithm: Penalized Power Iteration

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- For notational convenience, we define $S(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{m \times m}$, which simplifies expressions involving regularization. The penalty matrix Ω is set up so that larger values of the quadratic form $v^\top \Omega v$ mean rougher functions. This means that it penalizes functions that change quickly between time points.

Tuning Smoothness Parameters

- To select the optimal tuning parameters α efficiently, one can use a traditional Cross-Validation (CV) criterion and a computationally efficient closed-form Generalized Cross-Validation (GCV) criterion:

$$\text{CV}(\alpha) = \frac{1}{m} \sum_{j=1}^m \frac{\left[\{(I - S(\alpha))(X^T u)\}_{jj} \right]^2}{(1 - \{S(\alpha)\}_{jj})^2},$$

where $\{\cdot\}_{jj}$ denotes the j -th diagonal element.

$$\text{GCV}(\alpha) = \frac{1}{m} \frac{\|(I - S(\alpha))(X^T u)\|^2}{\left(1 - \frac{1}{m} \text{tr}\{S(\alpha)\}\right)^2}.$$

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Sparsity acts as built-in feature selection, focusing on key time points or locations.
- Sparse FPCA formulation [Shen and Huang, 2008]:

$$\min_{u,v} \|X - uv^T\|_F^2 + p_\gamma(v)$$

where $p_\gamma(v)$ is a sparsity-inducing penalty.

Sparsity penalties

- **Lasso (Soft-thresholding):**

$$p_{\gamma}^{\text{soft}}(|\theta|) = 2\gamma|\theta|, \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{soft}}(y) = \text{sign}(y)(|y| - \gamma)_+$$

- **Hard thresholding:**

$$p_{\gamma}^{\text{hard}}(|\theta|) = \gamma^2 I(|\theta| \neq 0), \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{hard}}(y) = I(|y| > \gamma) y$$

- **SCAD penalty:**

$$p_{\gamma}^{\text{SCAD}}(|\theta|) = \begin{cases} 2\gamma|\theta|, & |\theta| \leq \gamma, \\ \frac{\theta^2 - 2a\gamma|\theta| + \gamma^2}{a-1}, & \gamma < |\theta| \leq a\gamma, \\ \frac{(a+1)\gamma^2}{2}, & |\theta| > a\gamma, \end{cases} \xrightarrow{\text{minimizer}} h_{\gamma}^{\text{SCAD}}(y) = \begin{cases} \text{sign}(y)(|y| - \gamma)_+, & |y| \leq 2\gamma, \\ \frac{(a-1)y - \text{sign}(y)a\gamma}{a-2}, & 2\gamma < |y| \leq a\gamma, \\ y, & |y| > a\gamma, \end{cases}$$

where $a = 3.7$ (Fan and Li [2001]).

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 - ➌ **Normalize Right Singular Vector:** $v \leftarrow \frac{v}{\|v\|}$

Cross-Validation for Sparsity Selection

- **Sparsity parameter:** Tuning parameter controlling number of non-zero loadings in v ($0 =$ dense, $p =$ full sparsity).

Algorithm: K-fold CV Tuning Parameter Selection - Degree of sparsity

- 1 Randomly group the rows of side-by-side data matrix X into K roughly equal-sized groups, denoted as $X^1, \dots, X^K$.

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 - 2 Calculate the K-fold CV scores defined as: (N is the number of grid points in X^k)

$$CV_j = \sum_{k=1}^K \frac{\|X^k - u^{-k}(j)v^k(j)\|^2}{N}$$

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- 3 Select the degree of sparsity as $j_0 = \arg \min \{CV(j)\}$.

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- More stable & applicable to high-dimensional multivariate FDA

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- **Block structure example:**

$$X_i = \begin{bmatrix} x_{1i}(t_{i1}) & \cdots & x_{1i}(t_{i,m_i}) \\ \vdots & \ddots & \vdots \\ x_{ni}(t_{i1}) & \cdots & x_{ni}(t_{i,m_i}) \end{bmatrix}, \quad \mathbf{X} = [X_1 \ X_2 \ \cdots \ X_p]$$

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- The penalized reconstruction error is

$$\min_{u,v} \|\mathbf{X} - uv^\top\|_F^2 + \alpha^\top (v^\top \mathbf{\Omega} v),$$

where $\alpha = (\alpha_1, \dots, \alpha_p)^\top$ controls smoothness.

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- The smoothing operator is $\mathbf{S}(\alpha) = (I + \alpha\Omega)^{-1} \in \mathbb{R}^{M \times M}$.
 - The smoothing parameter α is selected via generalized cross-validation (GCV), defined as

$$\text{GCV}(\alpha) = \frac{1}{M} \frac{\|(I - \mathbf{S}(\alpha))(\mathbf{X}^\top u)\|^2}{\left(1 - \frac{1}{M}\text{tr}\{\mathbf{S}(\alpha)\}\right)^2}. \quad (1)$$

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- Algorithm CV Tuning for Sparsity and equation (1) are used to tune the sparsity level via K-fold CV and the smoothing parameter via GCV, respectively.

Simulation: Estimation Performance

- **Data-generating process:** Two functional variables:

$$X_{ij}^{(1)} = u_{i1}v_{11}(t_j) + u_{i2}v_{12}(t_j) + \epsilon_{ij}^{(1)}, \quad X_{ij}^{(2)} = u_{i1}v_{21}(t_j) + u_{i2}v_{22}(t_j) + \epsilon_{ij}^{(2)},$$

- where $u_{i1} \sim N(0, \sigma_1^2)$, $u_{i2} \sim N(0, \sigma_2^2)$, $\epsilon_{ij}^{(k)} \sim N(0, \sigma^2)$, and $n = m = 101$, $t_j \in [-1, 1]$

- **True functional PCs:**

- Variable 1: $v_{11}(t) = \frac{t + \sin(\pi t)}{s_1}, \quad v_{12}(t) = \frac{\cos(3\pi t)}{s_2}$

- Variable 2:

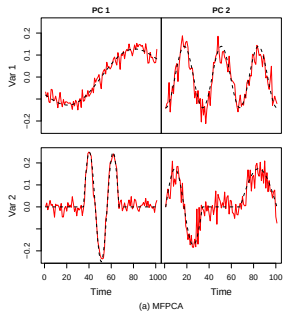
$$v_{21}(t) = \begin{cases} \frac{\sin(3\pi t)}{s_3}, & t \in (-\frac{1}{3}, \frac{1}{3}), \\ 0, & \text{otherwise,} \end{cases} \quad v_{22}(t) = \begin{cases} \frac{\sin(2\pi t)}{s_4}, & t \leq -\frac{1}{3}, \\ \frac{\sin(\pi t)}{s_4}, & t \geq \frac{1}{3}, \\ 0, & \text{otherwise.} \end{cases}$$

Here, s_1, s_2, s_3, s_4 are normalizing constants ensuring unit L^2 norm.

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Scenarios tested:

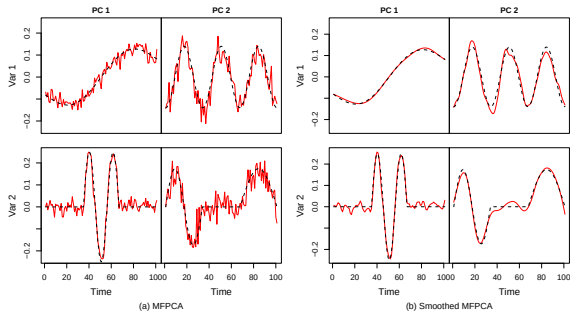
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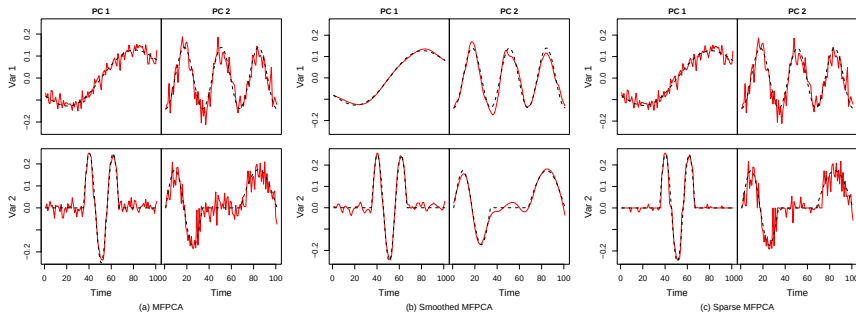
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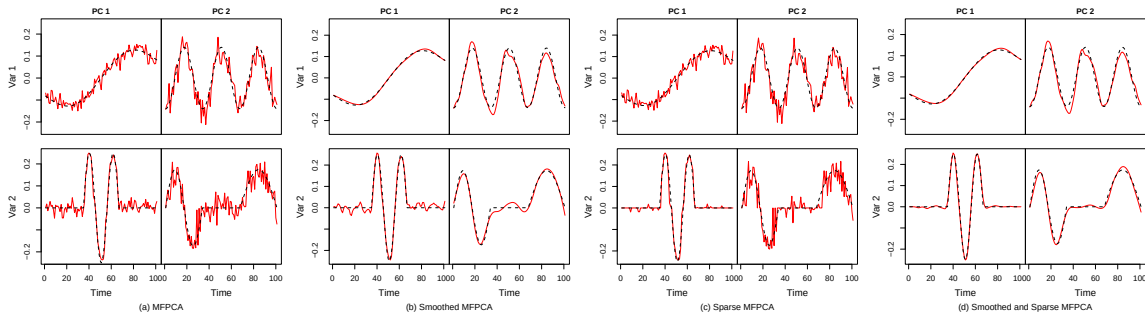
1. Unpenalized Multivariate SVD (baseline)
2. Smoothed Multivariate SVD (smoothness penalty)
3. Sparse Multivariate SVD (sparsity penalty)
4. Sparse + Smoothed Multivariate SVD (combined regularization)



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Accuracy measures:

- ① Variable-wise MSE:

$$\text{MSE}_{k\ell} = \frac{1}{m} \sum_{j=1}^m (\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

- ② Replication-averaged MSE:

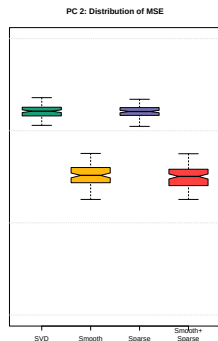
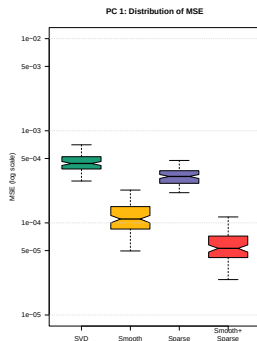
$$\overline{\text{MSE}}_{k\ell} = \frac{1}{R} \sum_{r=1}^R \text{MSE}_{k\ell}^{(r)}$$

- ③ Multivariate MSE:

$$\text{MSE}_{\ell}^{(\text{multi})} = \frac{1}{m} \sum_{j=1}^m \sum_{k=1}^2 (\hat{v}_{k\ell}(t_j) - v_{k\ell}(t_j))^2$$

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- **Performance across four methods (SVD, Smooth, Sparse, Smooth+Sparse):**
 - Smoothness and/or sparsity **reduce MSE** compared to unregularized SVD.
 - **Smooth+Sparse** yields lowest error and most stable estimates.
 - Smooth estimator performs consistently well; sparsity alone less effective (esp. for PC2).
 - Joint regularization achieves best **bias–variance tradeoff**.



PC1: Quartiles and Mean log₁₀(MSE)

Method	Q1	Median	Mean	Q3
SVD	-3.41	-3.35	-3.34	-3.28
Smooth	-4.07	-3.96	-3.92	-3.82
Sparse	-3.57	-3.50	-3.49	-3.43
Smooth+Sparse	-4.38	-4.28	-4.22	-4.14

PC2: Quartiles and Mean log₁₀(MSE)

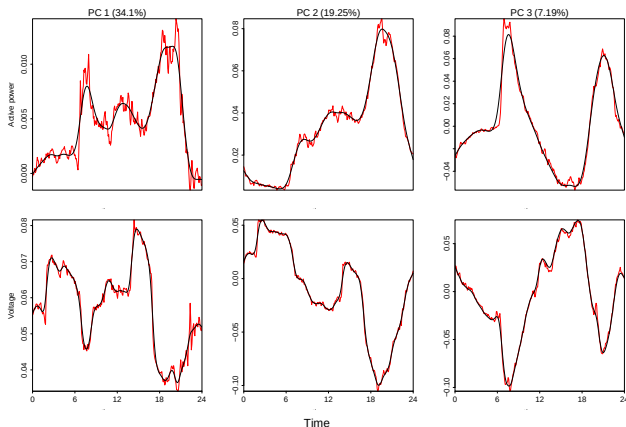
Method	Q1	Median	Mean	Q3
SVD	-2.84	-2.79	-2.79	-2.75
Smooth	-3.56	-3.48	-3.47	-3.40
Sparse	-2.83	-2.79	-2.79	-2.75
Smooth+Sparse	-3.59	-3.50	-3.49	-3.42

Application: Household Power Consumption

- **Dataset:** Bivariate functional data including active power and voltage consumption [Hebrail and Berard, 2012] for one household between December 2006 and November 2010.
- **Scaling:** To equalize the contribution of each variable in the multivariate analysis, we rescale them following [Happ and Greven, 2018].

$$\tilde{X}_j(t_i) = \hat{w}_j^{1/2} X_j(t_i),$$

$$\hat{w}_j = \left(\frac{1}{m} \sum_{i=1}^m \widehat{\text{Var}}(X_j(t_i)) \right)^{-1}.$$



First 3 PCs: MFPCA (red) vs ReMFPCA (black)

Regularization reduces noise while preserving the dominant daily consumption patterns, enhancing interpretability without losing key structure.

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 - Result: Low-rank, interpretable, noise-robust principal components for high-dimensional applications.

Two-way Smoothed MFPCA: Setup & Penalty

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- Multivariate v : $\mathbf{\Omega}_v = \text{blockdiag}(\mathbf{\Omega}_1, \dots, \mathbf{\Omega}_p).$

Two-way Smoothed MFPCA: Conditional GCV

- Minimizers:

$$u = \frac{\mathbf{S}_u(\alpha_u) \mathbf{X}_v}{v^\top (I + \alpha_v \mathbf{\Omega}_v) v} = \frac{\mathbf{S}_u(\alpha_u)}{1 + \alpha_v \mathbf{R}_v(v)} \frac{\mathbf{X}_v}{\|v\|^2}, \quad v = \frac{\mathbf{S}_v(\alpha_v) \mathbf{X}^\top u}{u^\top (I + \alpha_u \mathbf{\Omega}_u) u} = \frac{\mathbf{S}_v(\alpha_v)}{1 + \alpha_u \mathbf{R}_u(u)} \frac{\mathbf{X}^\top u}{\|u\|^2}.$$

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- Optimization:** Alternate updates of u and v using GCV until convergence $\rightarrow$ two-way regularized components.

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- **Result:** Robust, interpretable, noise-resistant components for high-dimensional data.

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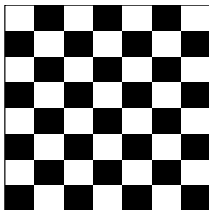
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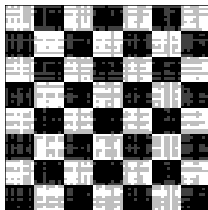
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 - ➋ $v \leftarrow \mathbf{S}_v^{[\alpha_v]} \mathbf{h}_v^{[\gamma_v]}(\mathbf{X}^\top u)$
 - ➌ $v \leftarrow v / \|v\|$
- ➌ $\mathbf{X} \leftarrow \mathbf{X} - \sigma u v^\top$ to extract the next component.

Chess Board

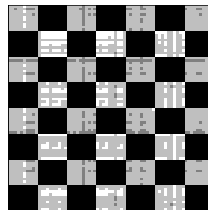
True Chessboard



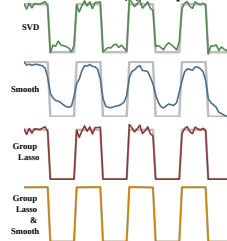
PCA with SVD



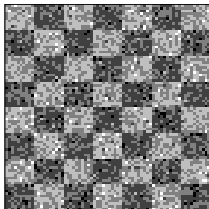
PCA with Group Lasso Penalty



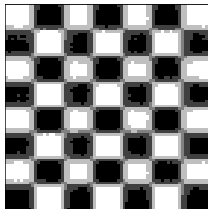
First PC Scores (u) Comparison



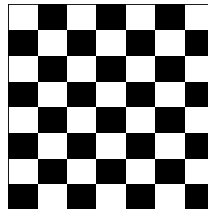
Noisy Chessboard



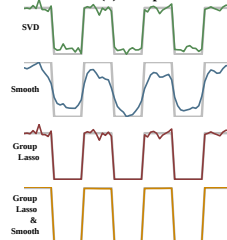
PCA with Smoothness Penalty



PCA with Group Lasso and Smoothness Penalties



First PC (v) Comparison



Selection of Regularization Parameters

- Four sets of tuning parameters:
 - α_u : smoothness of u , γ_u : sparsity of u
 - α_v : smoothness of v , γ_v : sparsity of v
- **Challenge:** Ordering of tuning (smoothness vs. sparsity) affects convergence and solutions.
- **Strategy: Conditional tuning**
 - 1 Initialize all penalties at 0.
 - 2 Tune γ_u via K -fold CV.
 - 3 Sequentially tune $\gamma_{v,i}$ using Algorithm~??.
 - 4 With sparsity fixed, tune α_u by GCV.
 - 5 Tune $\alpha_{v,i}$ using two-way GCV.
 - 6 Iterate steps 2–5 until stable.
- This alternating scheme **isolates sparsity vs. smoothness** while ensuring accuracy + interpretability.

K-Fold CV algorithm for Sparsity

K-Fold CV (Row Sparsity)

- 1 Split $\mathbf{X} \in \mathbb{R}^{n \times M}$ into K column groups $\{\mathbf{X}^{(1)}, \dots, \mathbf{X}^{(K)}\}$.

K-Fold CV + 1-SE Rule

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Variance Explained: Classical vs Regularized FPCA

- **Classical FPCA:** Loadings v_j orthonormal; scores $u_j = Xv_j$ uncorrelated. Variance explained by first J PCs:

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- **Issue under regularization:** smoothness/sparsity break orthogonality $\rightarrow$ scores become correlated $\rightarrow$ naive sum $\sum \|u_j\|^2$ **double-counts** variance (cf. Huang et al., 2008).

Subspace-Projection Definition of Explained Variance

- Normalize loadings and stack:

$$V_J = [v_1, \dots, v_J], \quad v_j \leftarrow \frac{v_j}{\|v_j\|}.$$

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- Projected data and explained variance:

$$X_J = XH_J, \quad V_{\text{tot}} = \text{tr}(X^\top X), \quad \mathcal{V}_J = \|X_J\|_F^2 = \text{tr}(H_J X^\top X H_J).$$

PVE, Incremental PVE, and Properties

- Incremental variance:

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- **Key properties:**

- No double-counting (works with correlated scores).
- Reduces to classical PCA when $V_J^\top V_J = I_J$.
- Monotone in J ($\mathcal{V}_J$ increases).
- $\Delta \mathcal{V}_j$ = unique variance added by component j .

Conclusion & Future Work

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